

# Light-UP

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The project Light-Up develops a modular vision assistance system which operates on laptops to help blind and visually impaired users. The system uses real-time vision inference to create explainable power modeling. The system uses a FastAPI backend and Streamlit frontend to enable real-time sensing and to support testing which can be replicated. The system estimates energy and power consumption together with runtime across different mobile SoC profiles and operating modes while supporting future development of advanced vision-language reasoning capabilities.

## Introduction

Visually impaired users need assistive systems which enable them to understand their surroundings and operate their devices without needing to use their hands and which function properly even when power supply is limited. The conventional methods which exist today do not provide two essential functions which include system flexibility and power management abilities. This project addresses these challenges by:

- Designing a modular, extensible vision assistant
- Simulating power-aware behavior of wearable devices
- Supporting real-time object, face, and scene understanding
- Preparing the system for Vision-Language Model (VLM) integration

## Methods

The Light-UP system was created through the combination of modular elements and power-efficient design methods. The development of each module started with its independent design process which established specific interface requirements that allowed for simultaneous work and future development.

- **Camera & Input Module:** Captures live video or injected streams and adapts frame rate and resolution based on power mode.
- **Vision Processing Module:** Performs object detection, face recognition, and scene description in independent processing modes.
- **Vision-Language Module:** Enables contextual reasoning and task-specific responses through natural language queries.
- **Audio & Voice Module:** Provides spoken feedback and offline voice command recognition for hands-free interaction.
- **Power Monitoring Module:** Simulates battery state and dynamically switches between Normal, Balanced, and Critical modes.
- **Mobile SoC Simulation Module:** Estimates energy consumption, power usage, and battery runtime for modern mobile processors.
- **System Integration Module:** Coordinates all modules, manages system state, and ensures stable real-time operation.

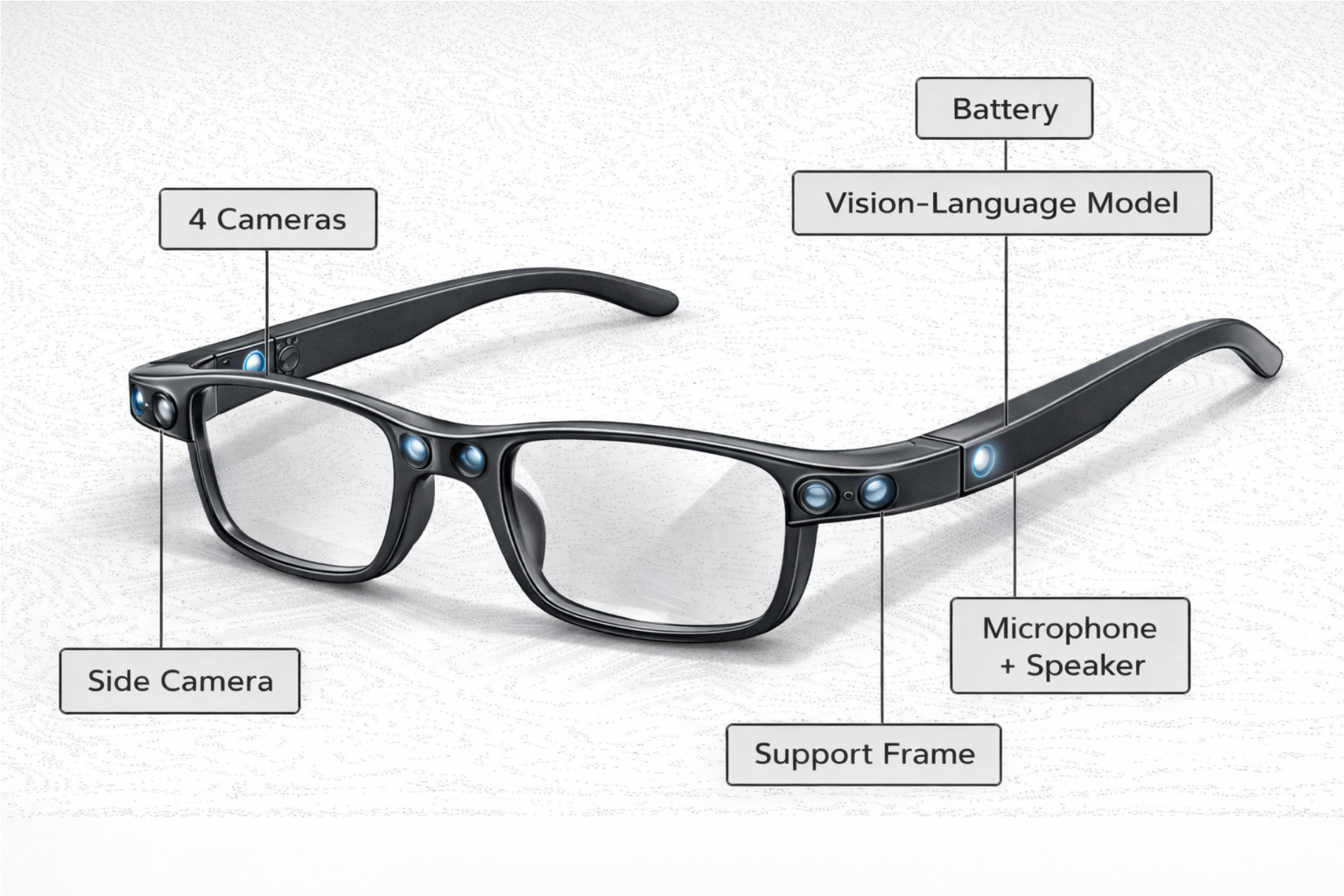
## Results

- The power simulation produces consistent and realistic energy estimates across all evaluated mobile SoC profiles.
- Balanced power policy achieves an effective trade-off between performance and battery runtime for real-time object detection.
- Battery-oriented policy significantly extends runtime while reducing average power consumption.
- Performance-oriented policy minimizes latency but increases energy usage and reduces operational duration.
- Estimated runtimes and power values align with reported benchmarks for modern flagship smartphones.

| SoC                | Policy      | Energy (mj) | Avg Power (mW) | Runtime (h) | NPU (%) |
|--------------------|-------------|-------------|----------------|-------------|---------|
| Snapdragon 8 Gen 3 | Balanced    | 48.74       | 1392.61        | 13.64       | 61.7    |
| Snapdragon 8 Gen 3 | Performance | 56.82       | 1623.43        | 11.70       | 73.2    |
| Snapdragon 8 Gen 3 | Battery     | 36.21       | 1034.57        | 18.36       | 42.8    |
| Dimensity 9300     | Balanced    | 51.33       | 1466.57        | 12.96       | 58.4    |
| A19-class          | Balanced    | 45.18       | 1290.86        | 14.72       | 64.3    |

Table 1: Estimated power consumption for object detection (25 ms inference, 4 ms decode, 6 ms post-processing)

## Outlook



**Figure:** Sleek smart-glasses prototype with four-camera system (two front-facing and one on each side), integrated microphone and speaker, battery, and Vision-Language Model embedded within the frame.

## Contact

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## Project Setting

Made by students in the bridge semester for the Master of Artificial Intelligence for industrial applications.